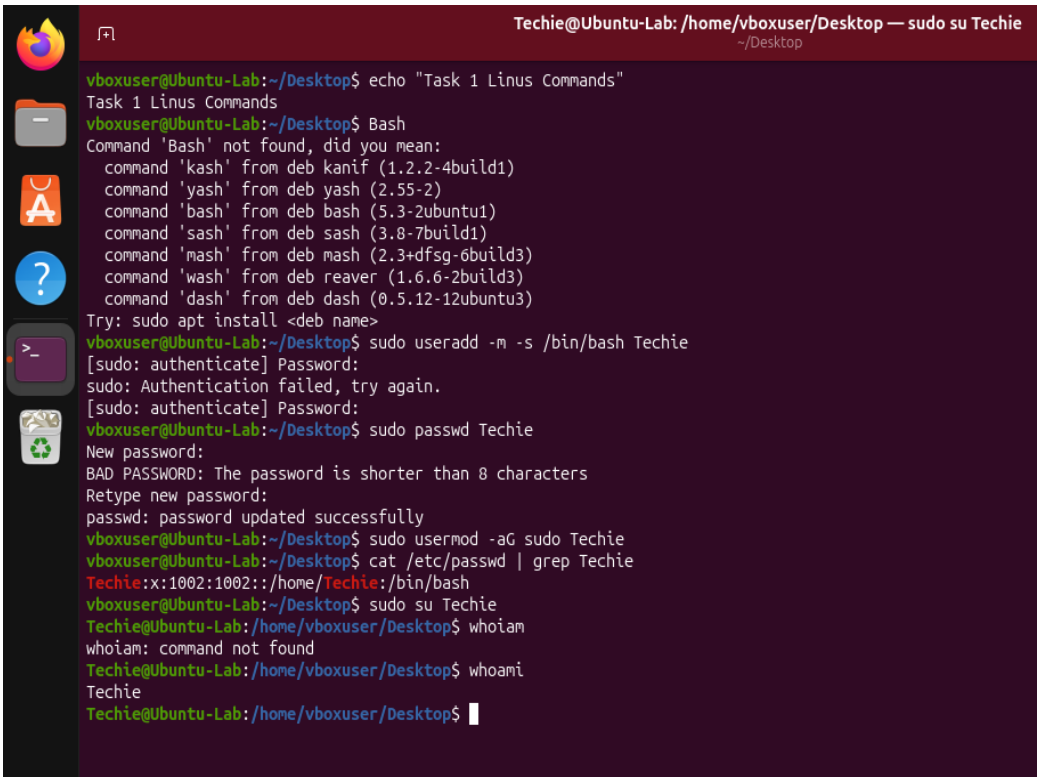
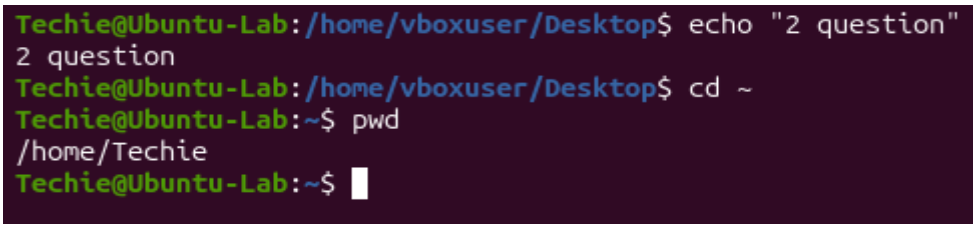
Task -1

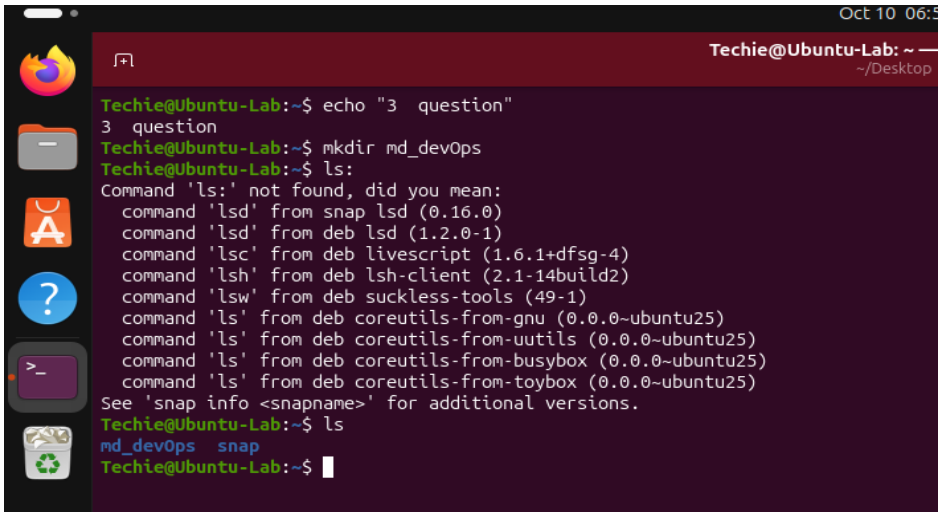
1. **Create user with name Techie and provide Sudo access to user.**



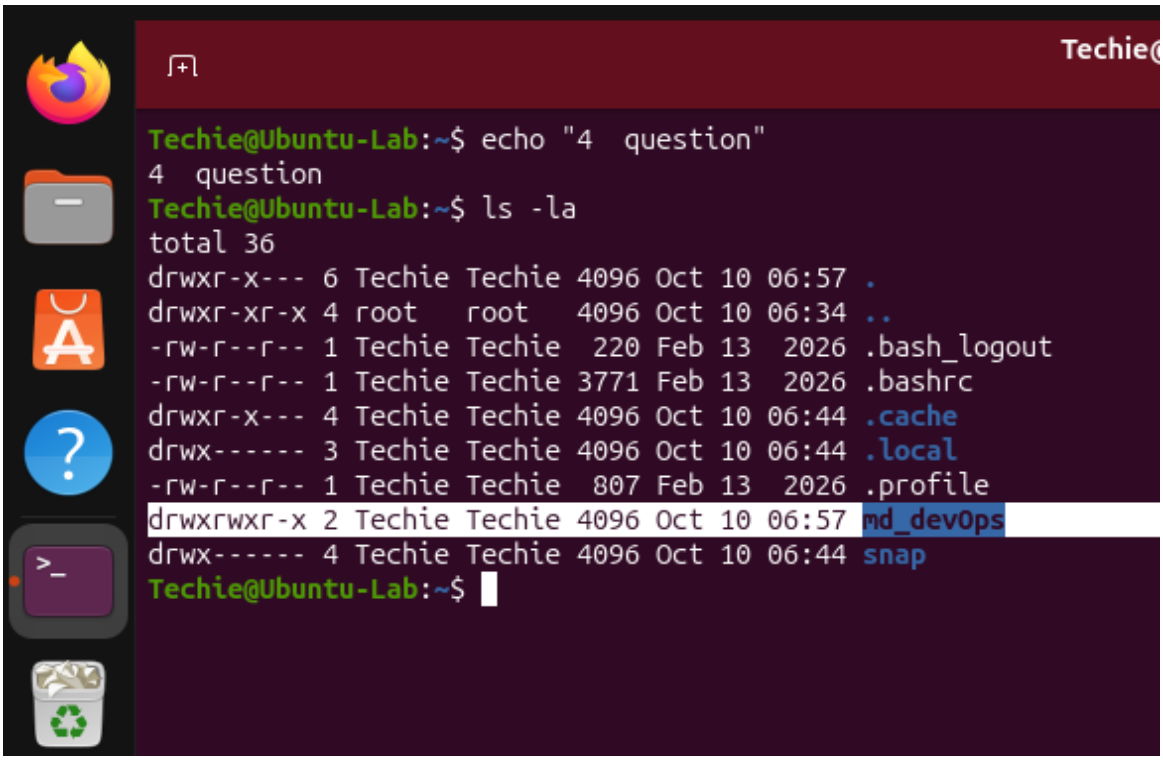
2.Navigate to the home directory.



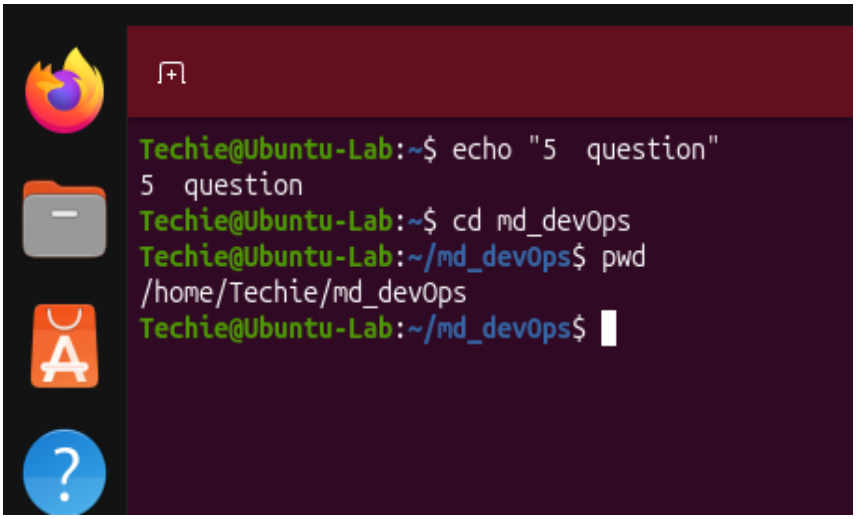
3. **Create a new directory.**



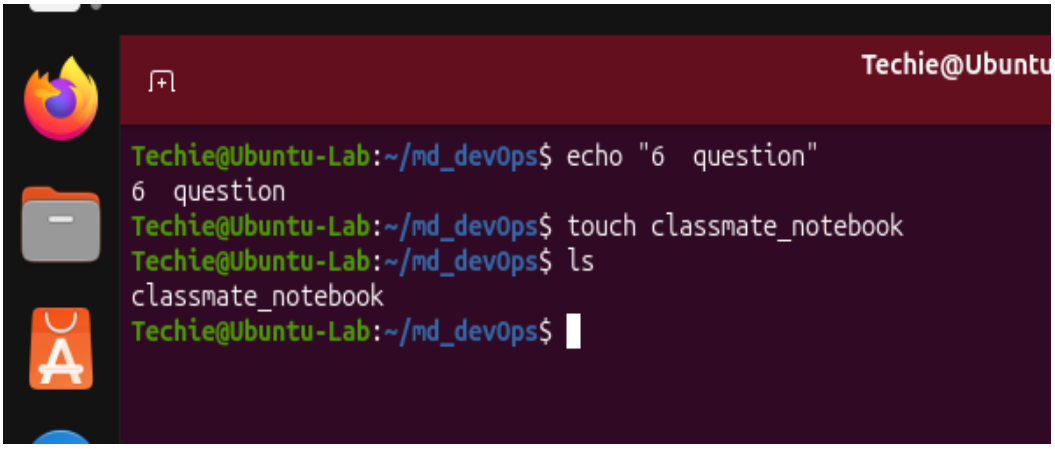
4. **List the contents of a directory.**



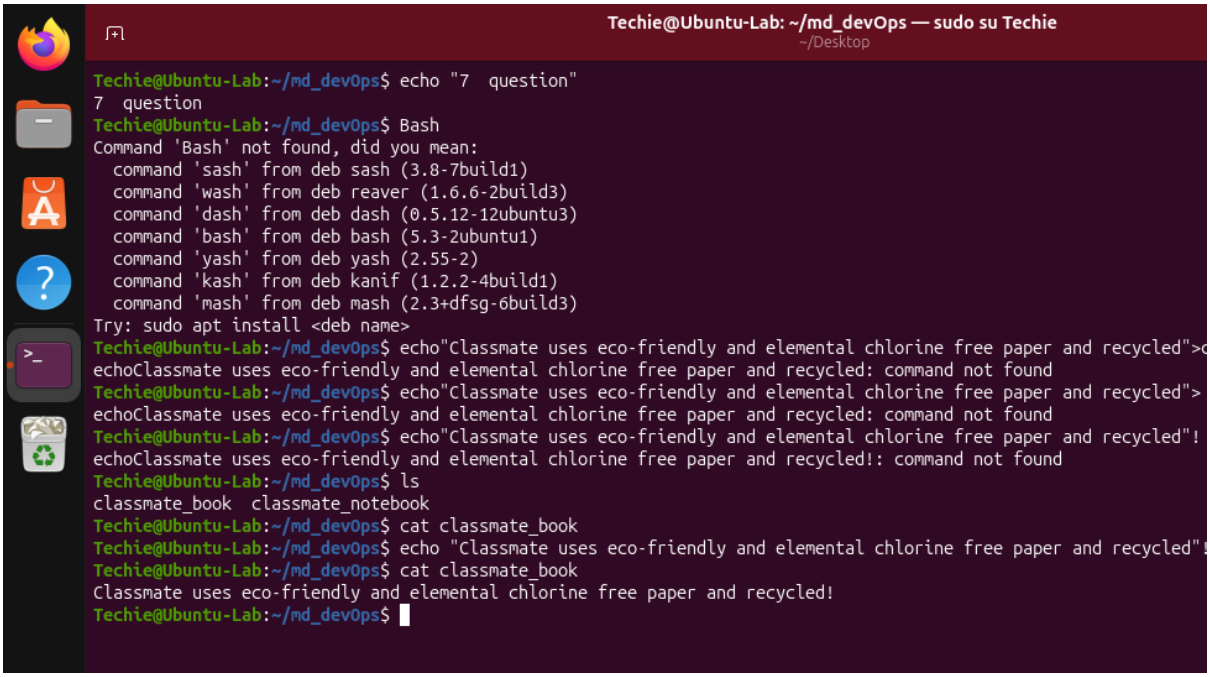
5. **Change the current directory.**



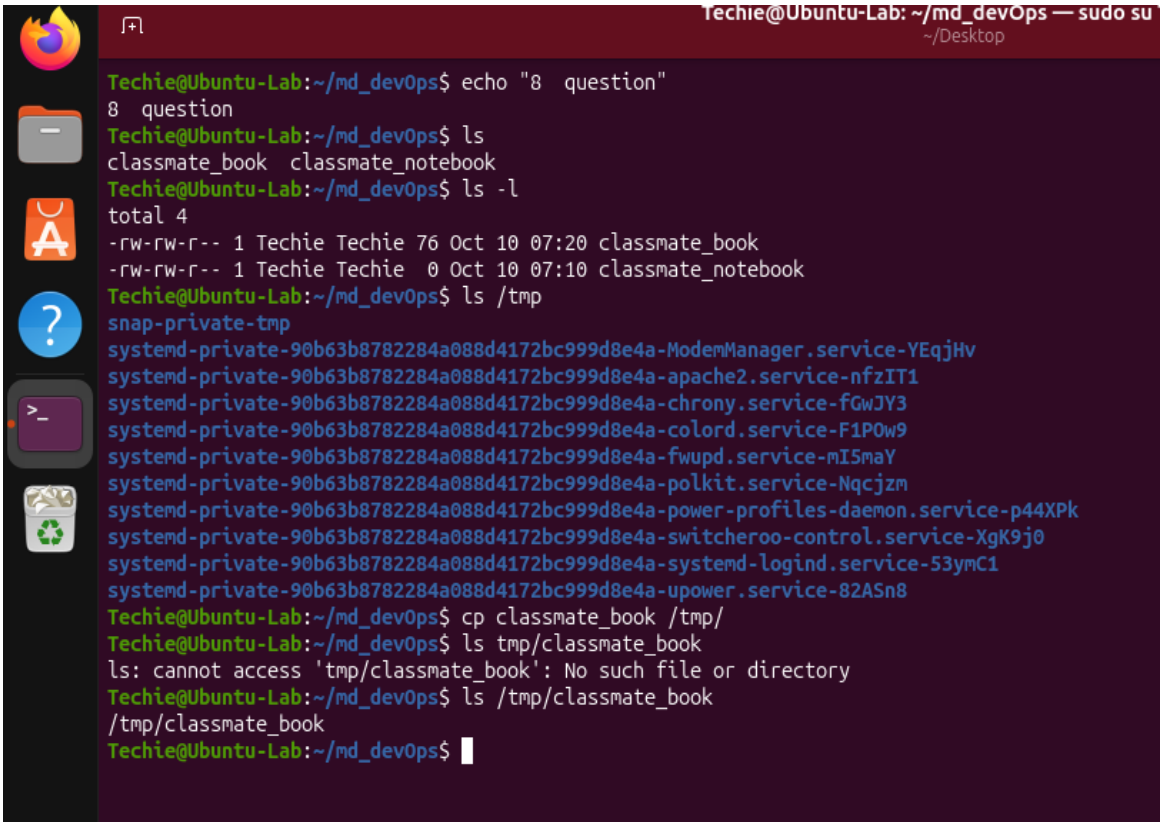
6. **Create a new empty file**



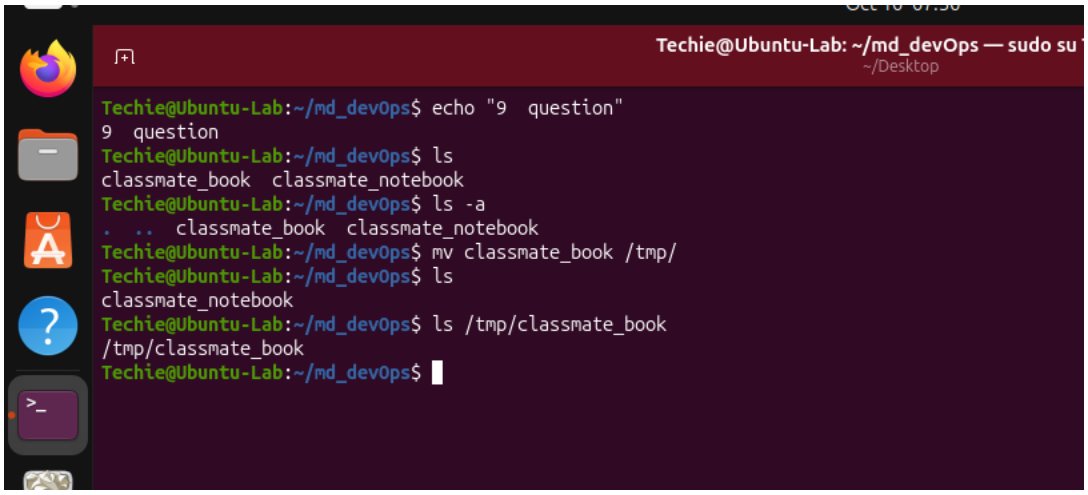
7. View the contents of a file.



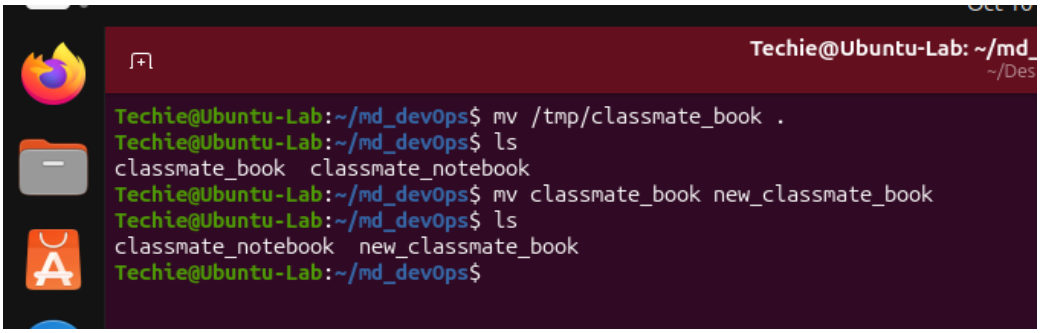
8. Copy a file to another location



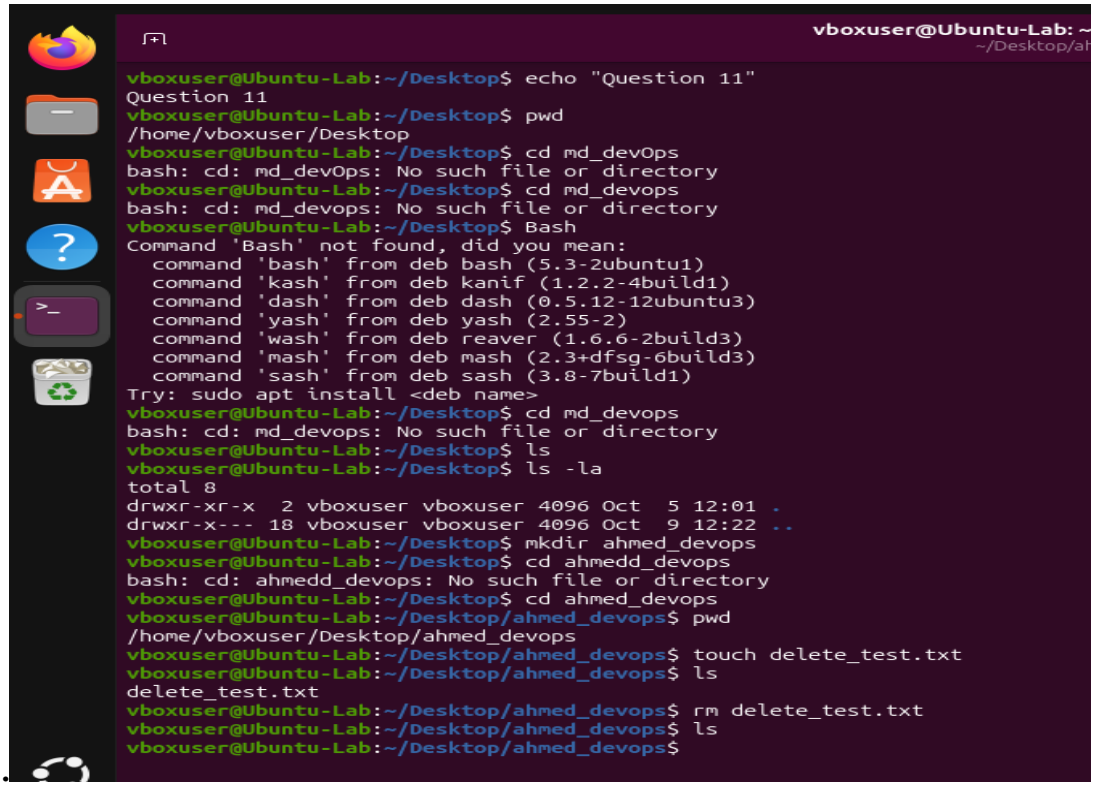
9. Move a file to another location.



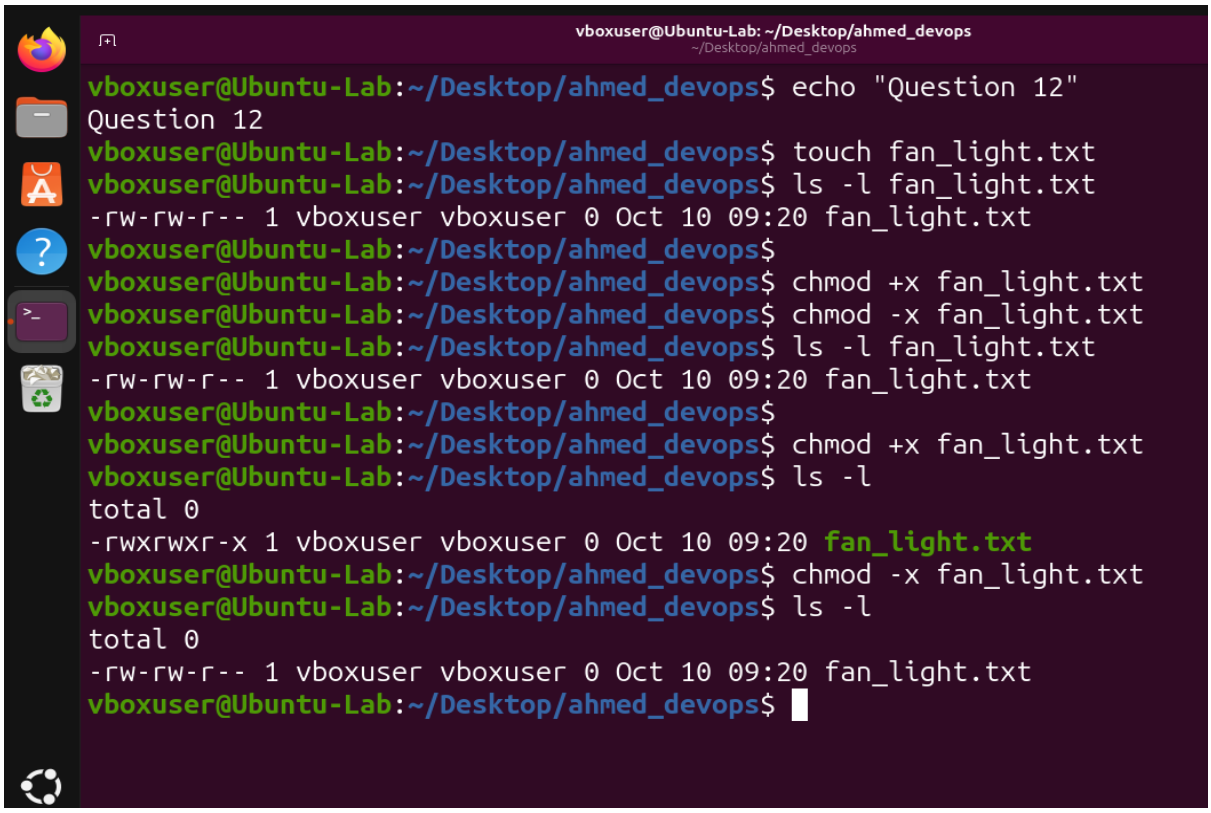
10. Rename a file.



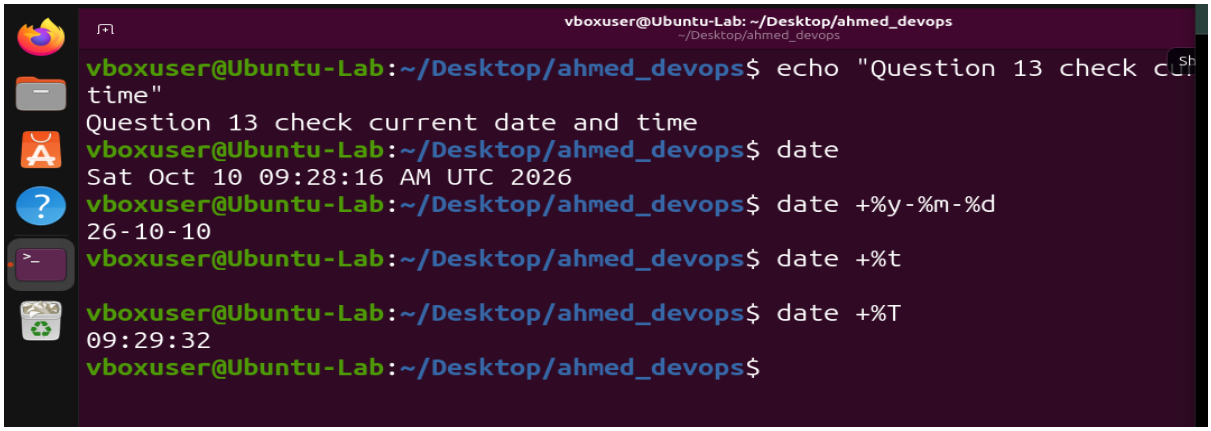
11.Delete the file



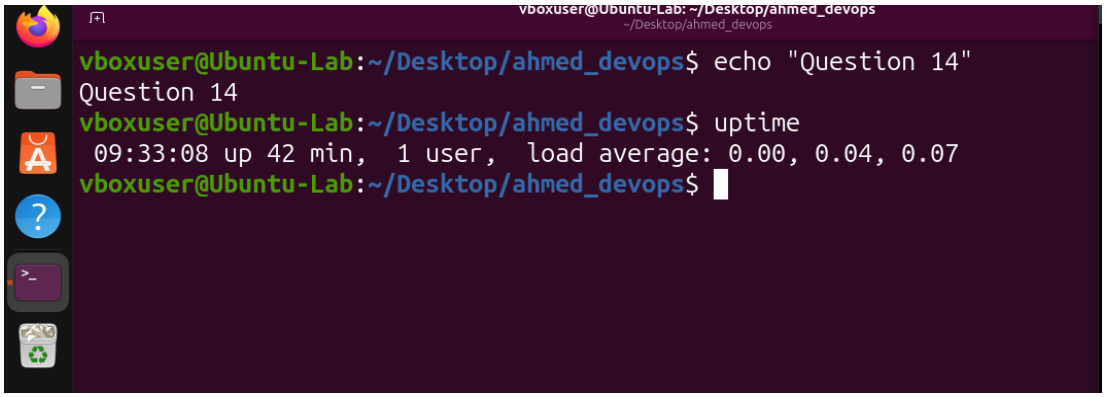
12.Grant or revoke permissions on a file or directory.



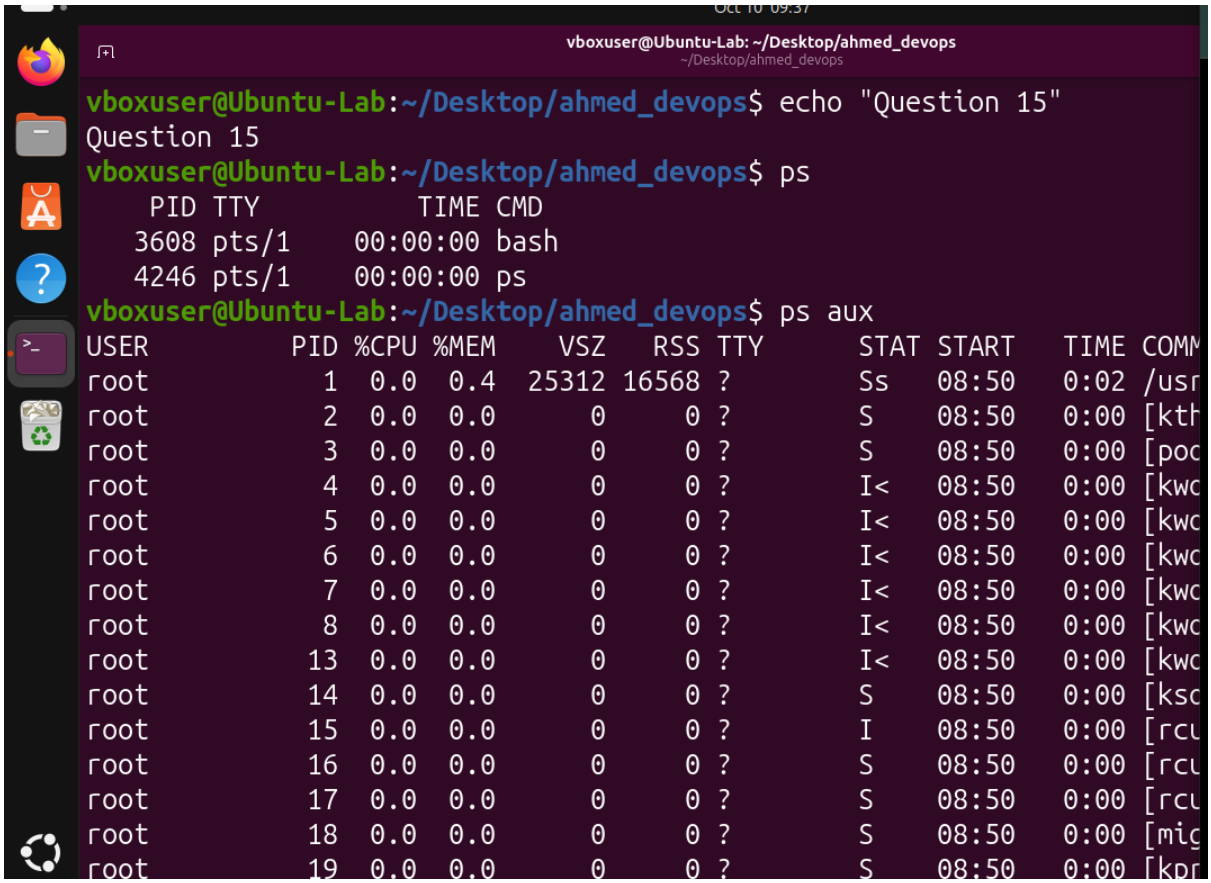
13. View the current date and time.



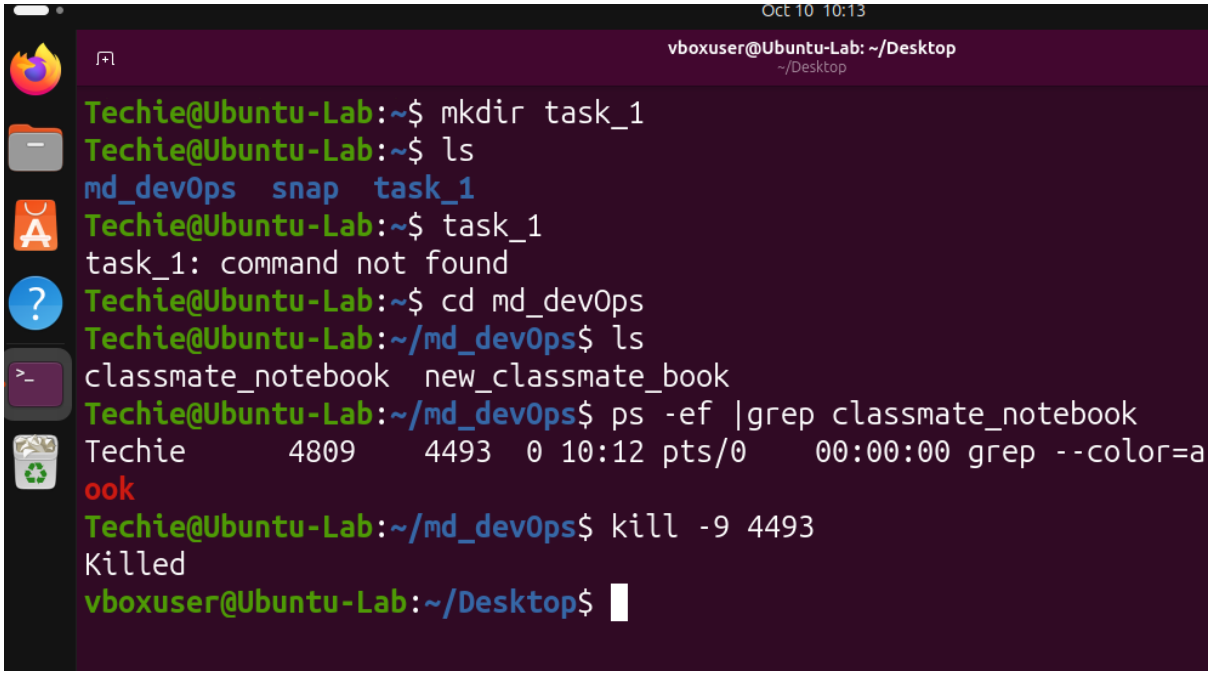
14. Check the system uptime.



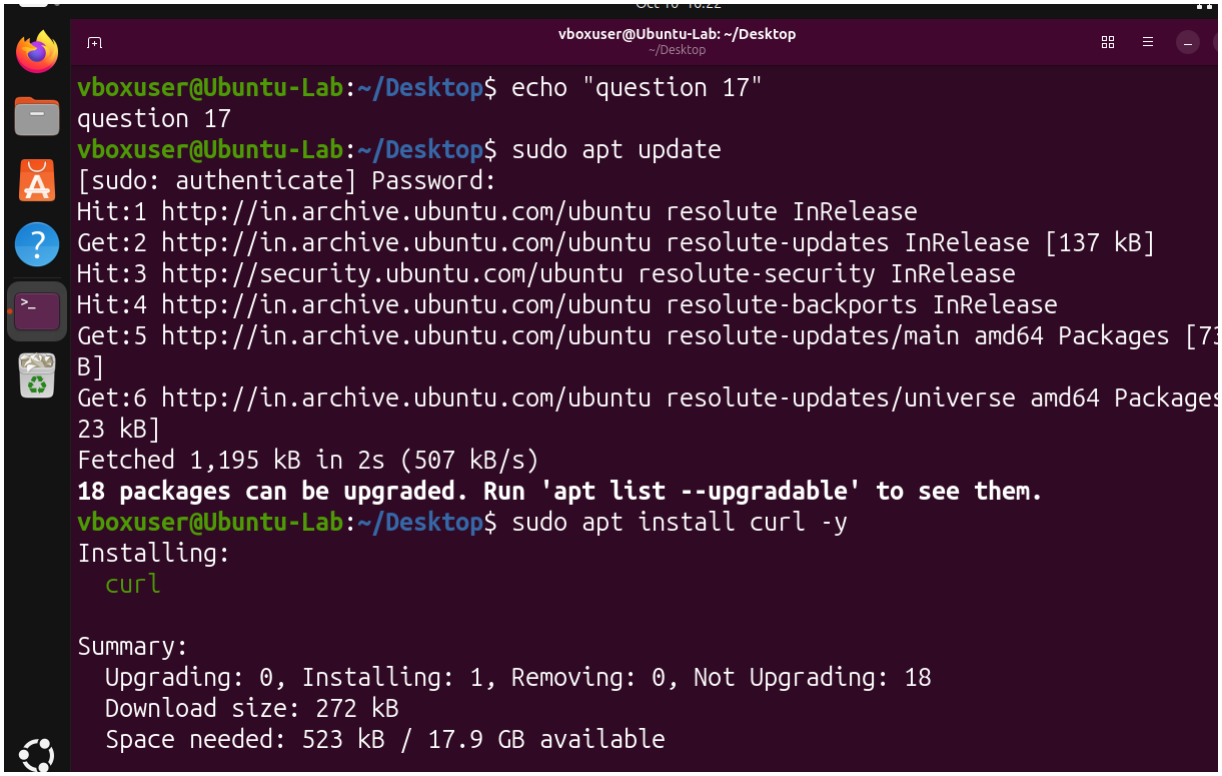
15. View the running processes.



16. Kill a running process.

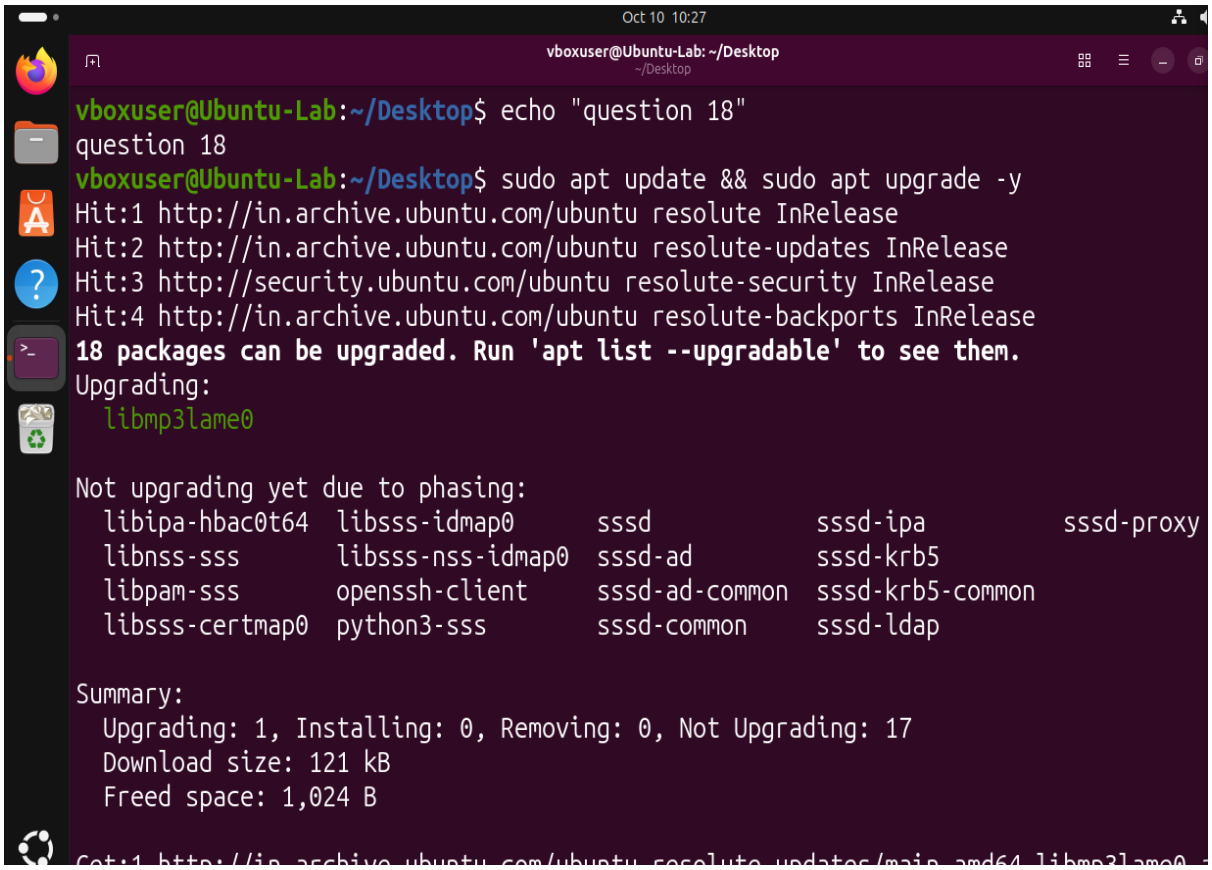


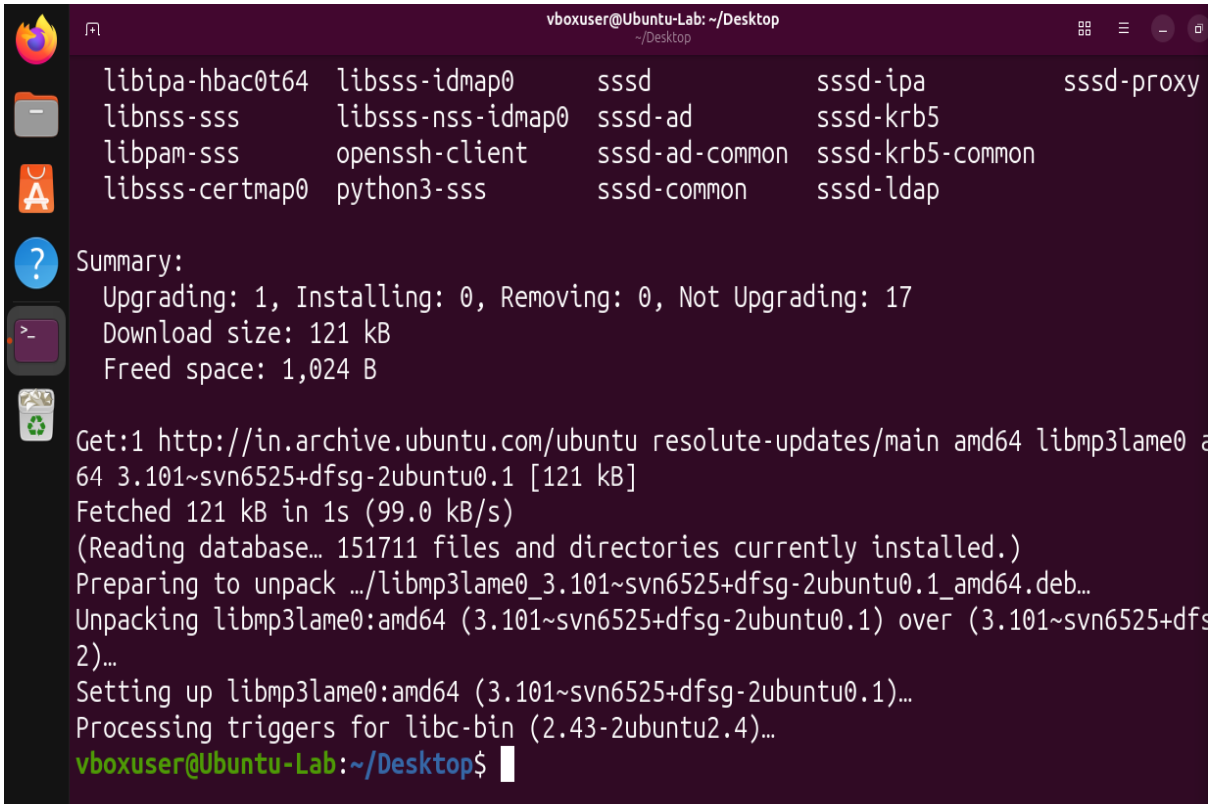
17. Install a package using the package manager (e.g., apt or yum).



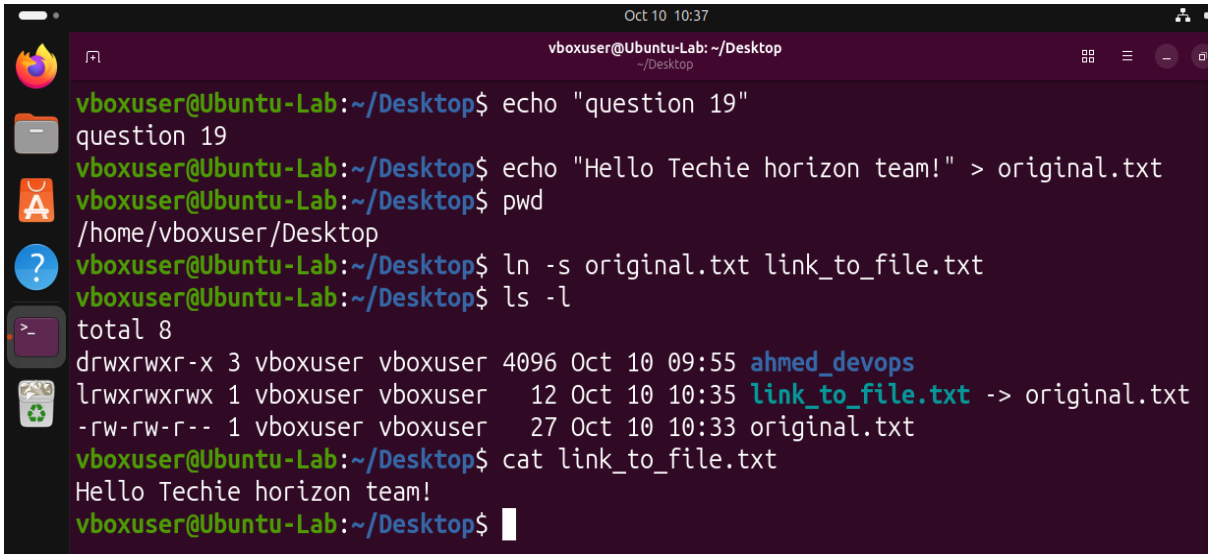


18. Update the system packages.

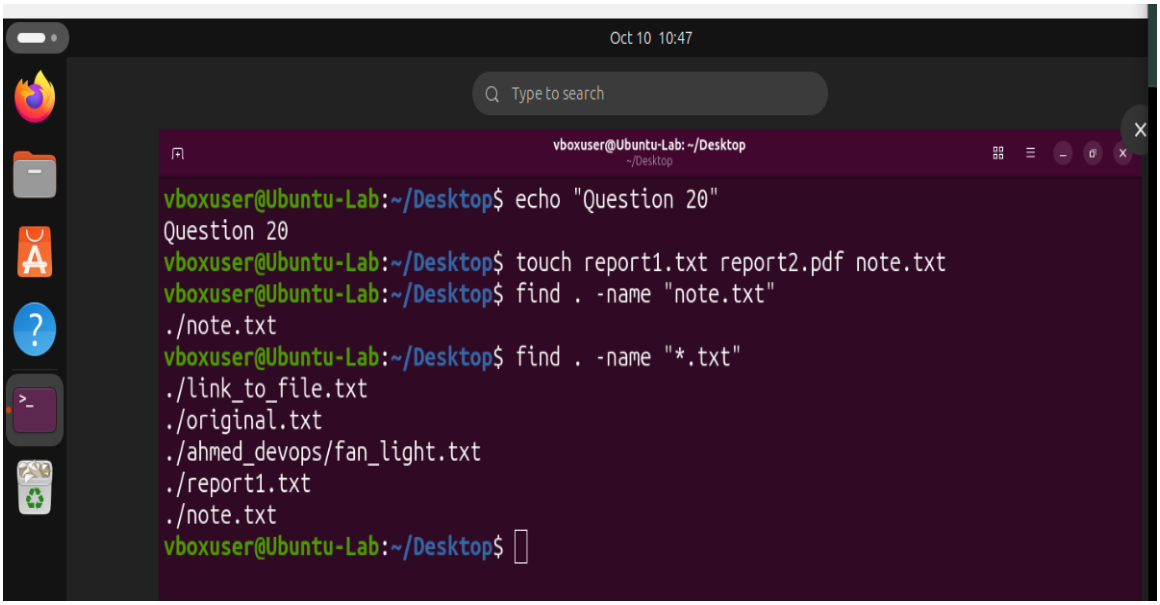




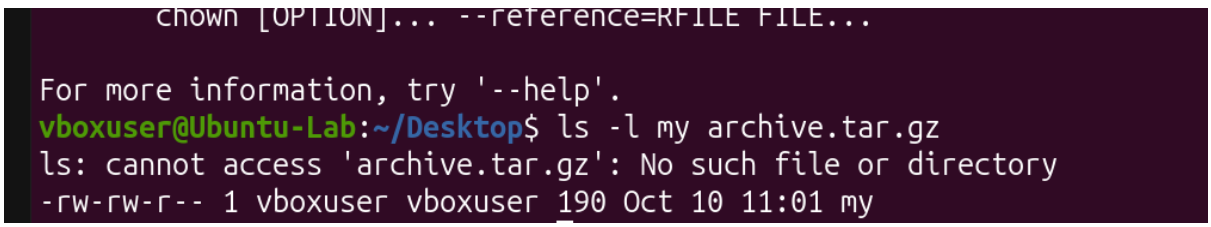
19. Create a symbolic link.



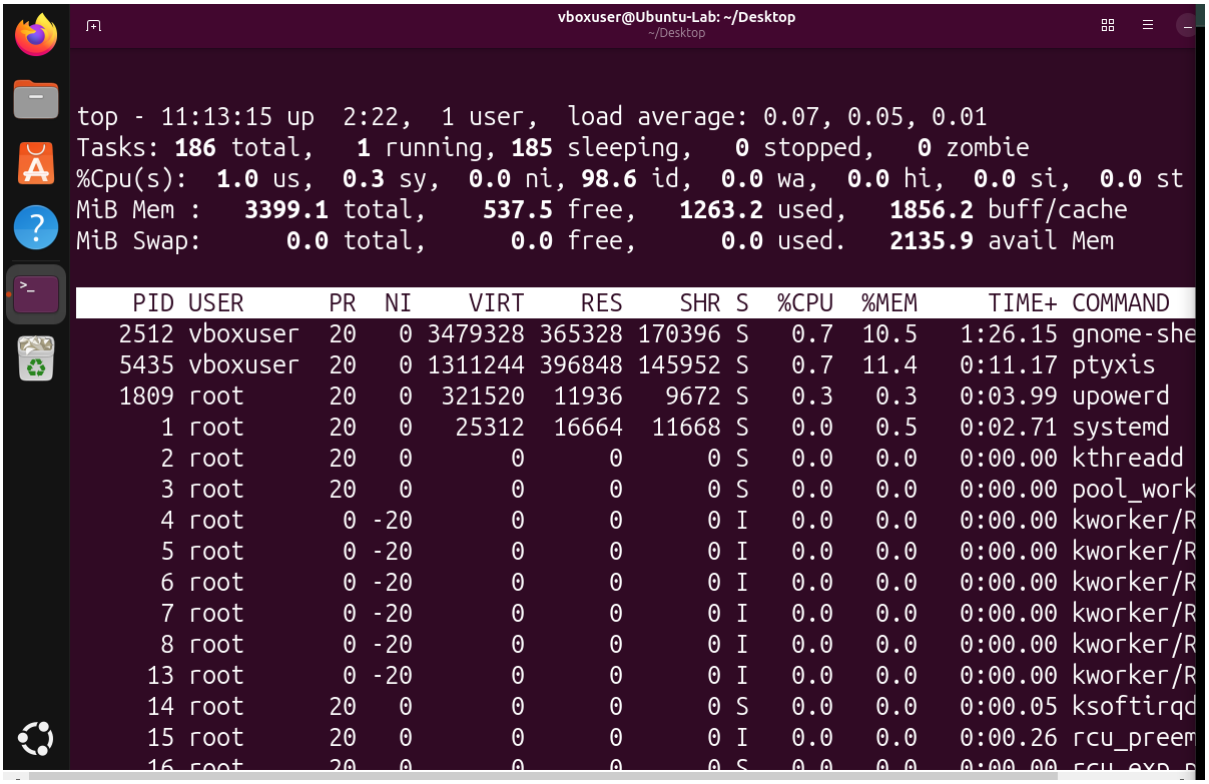
20. Search for files using the find command.



21. Compress and decompress files using tar.



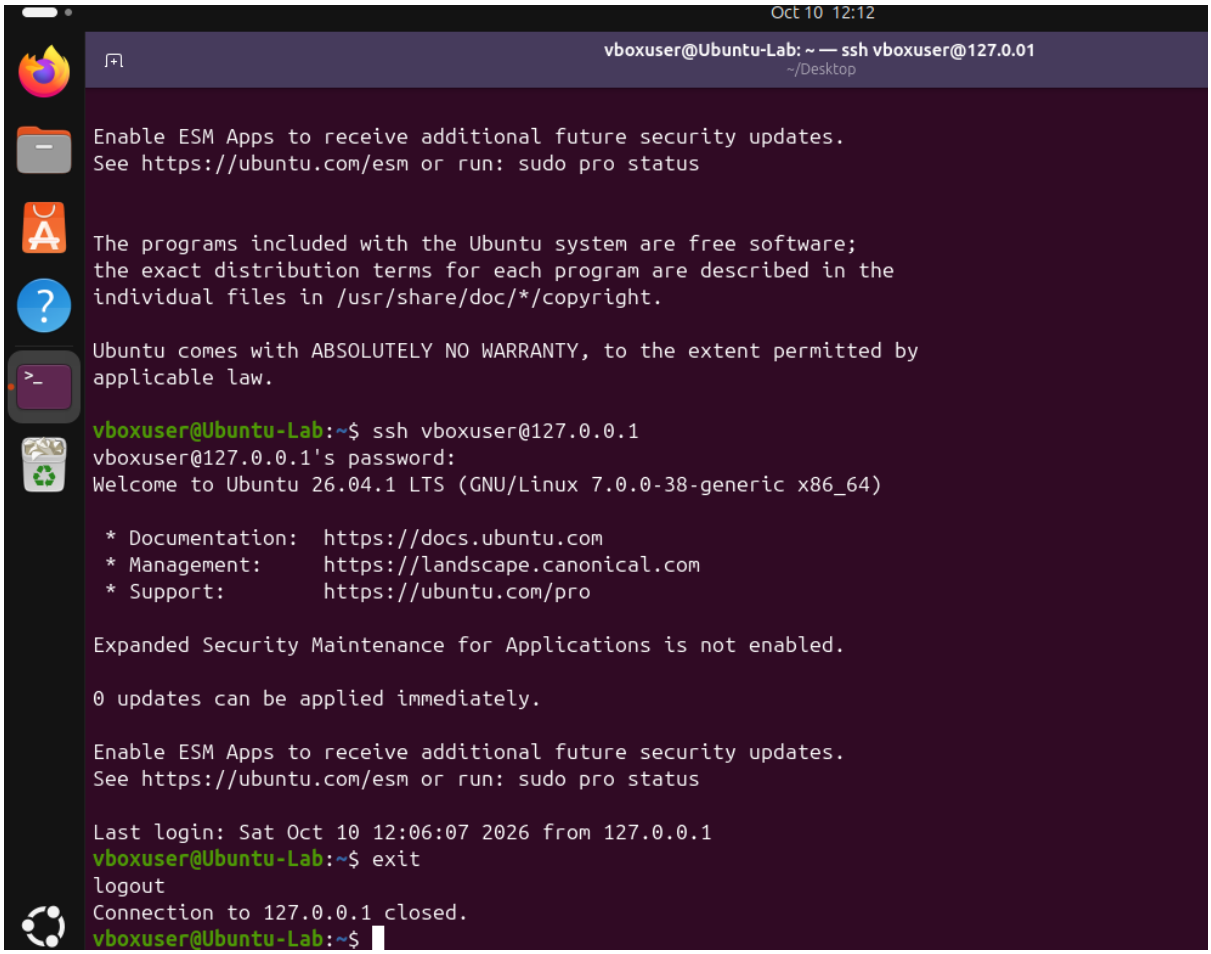
22. Monitor system resources with top or htop.



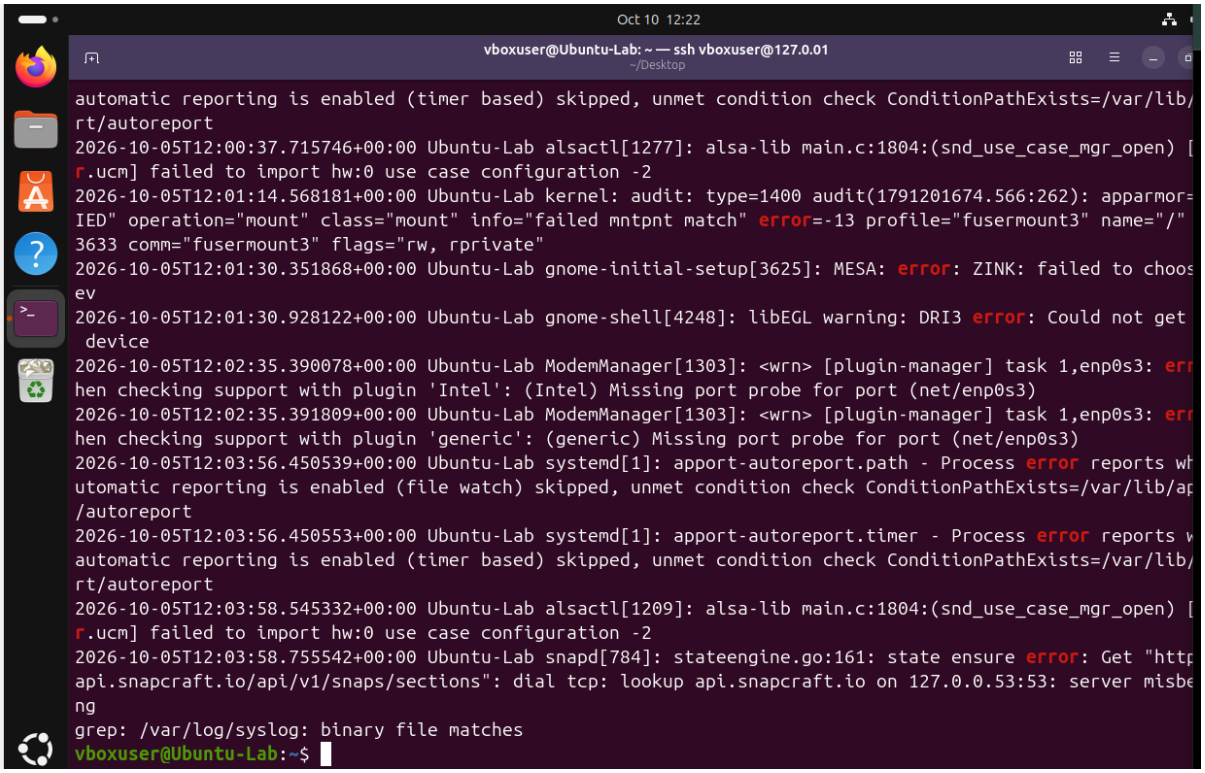
23. Create and manage user groups.



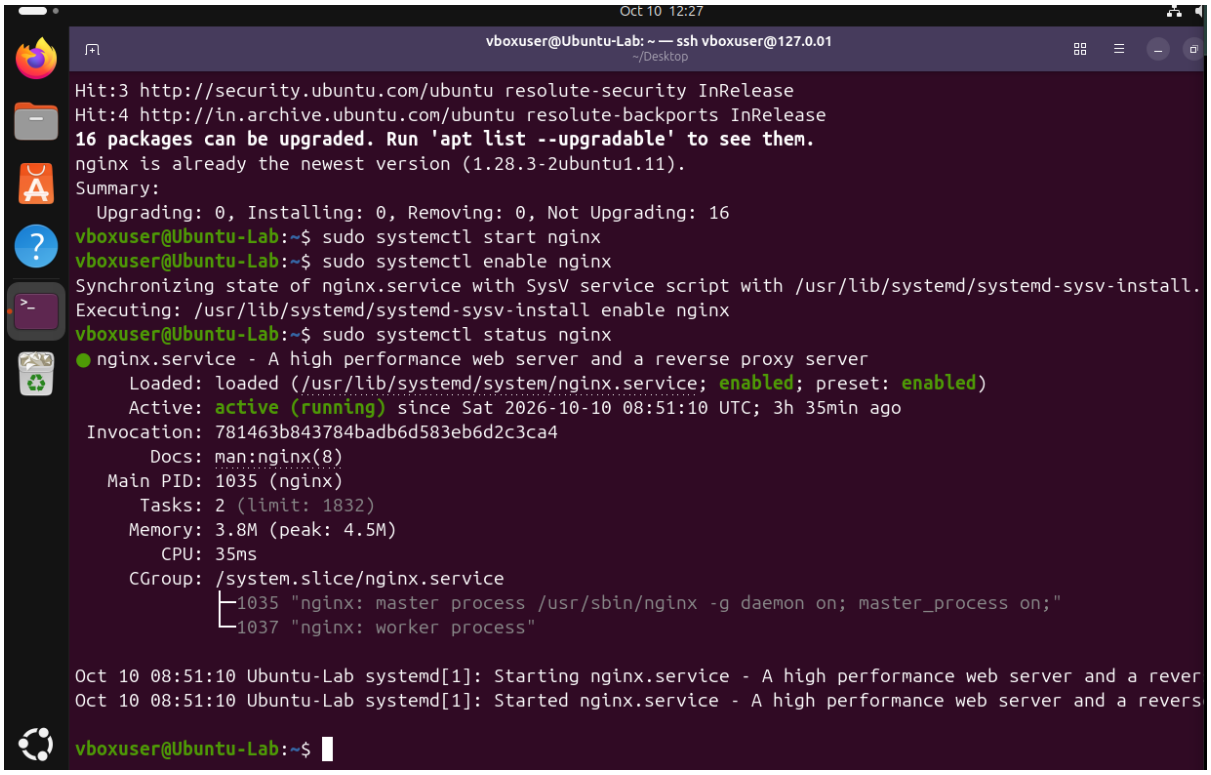
24. Set up SSH password less authentication.

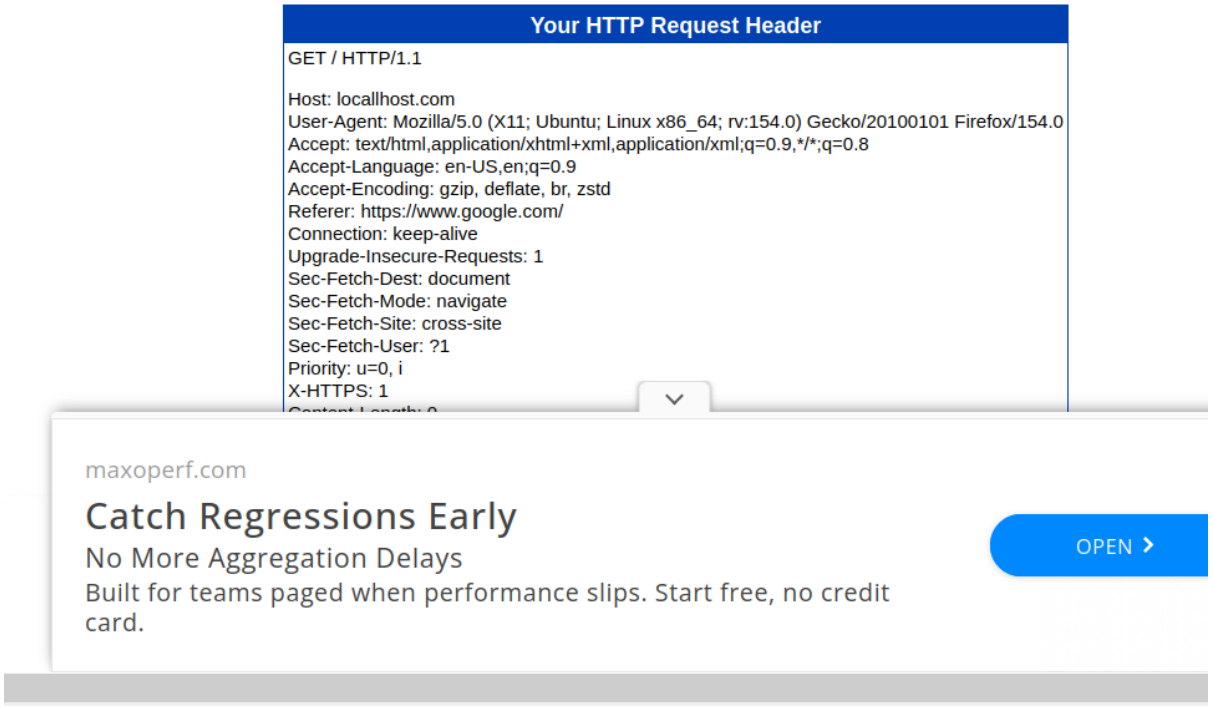


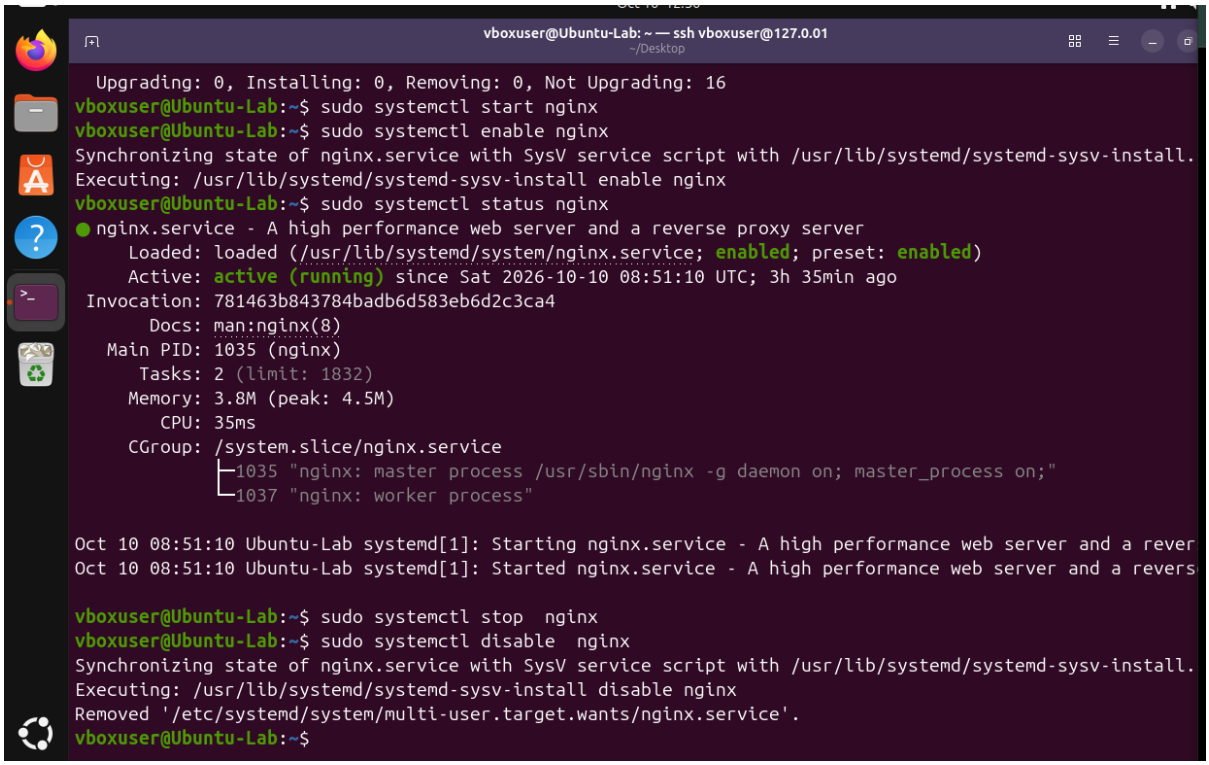
25. Monitor log files using tail or grep.



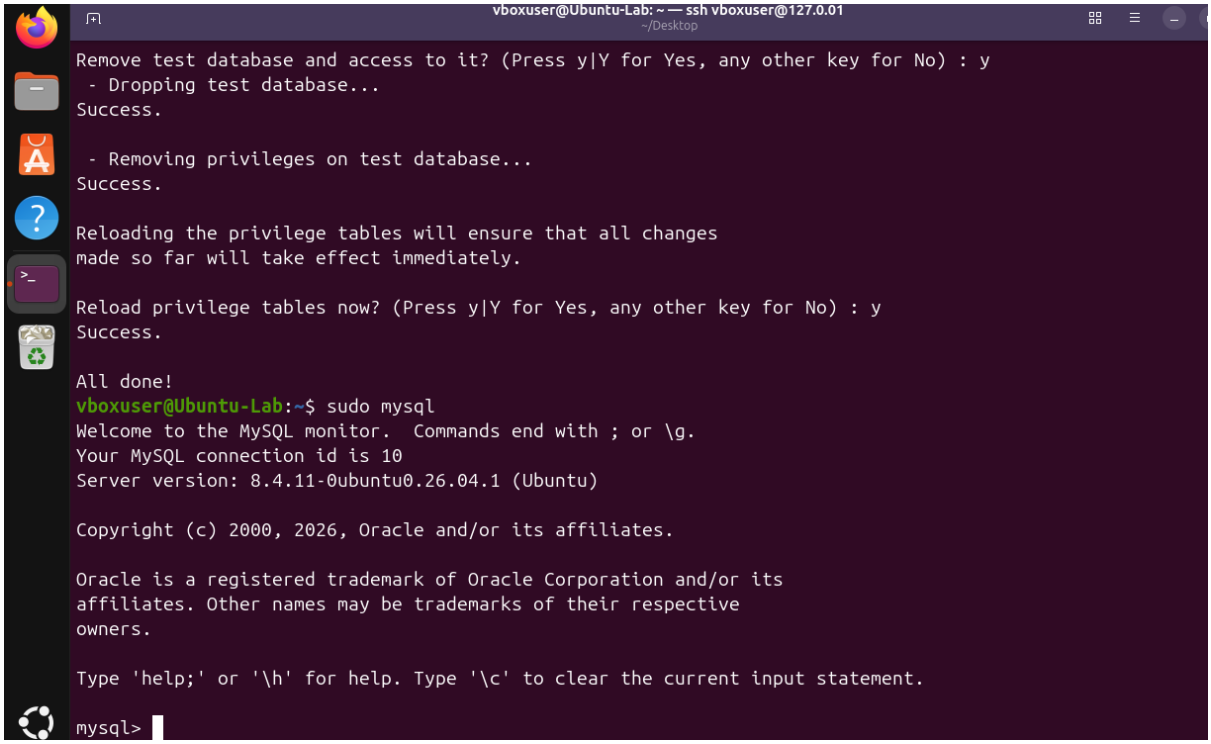
26. Set up a web server (e.g., Apache or Nginx).



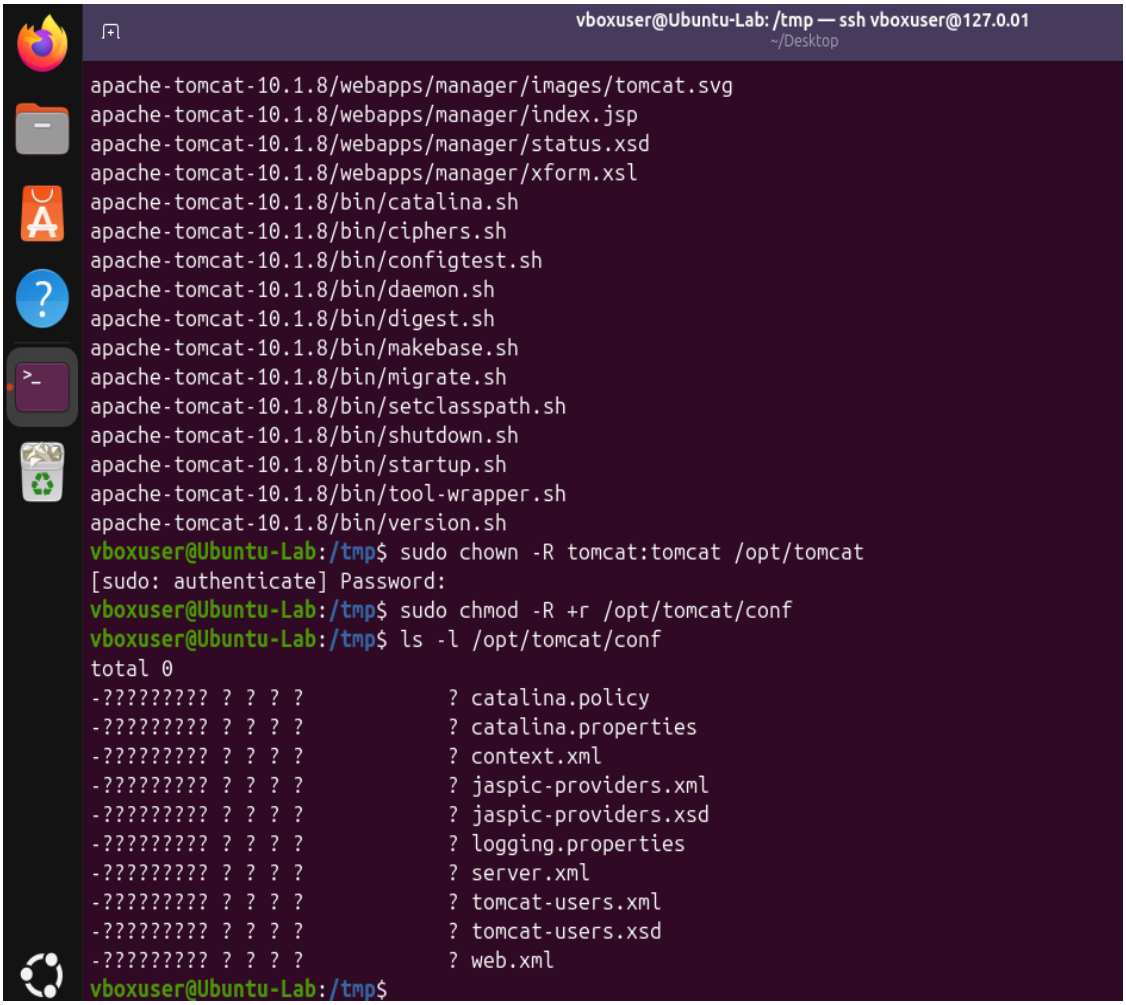




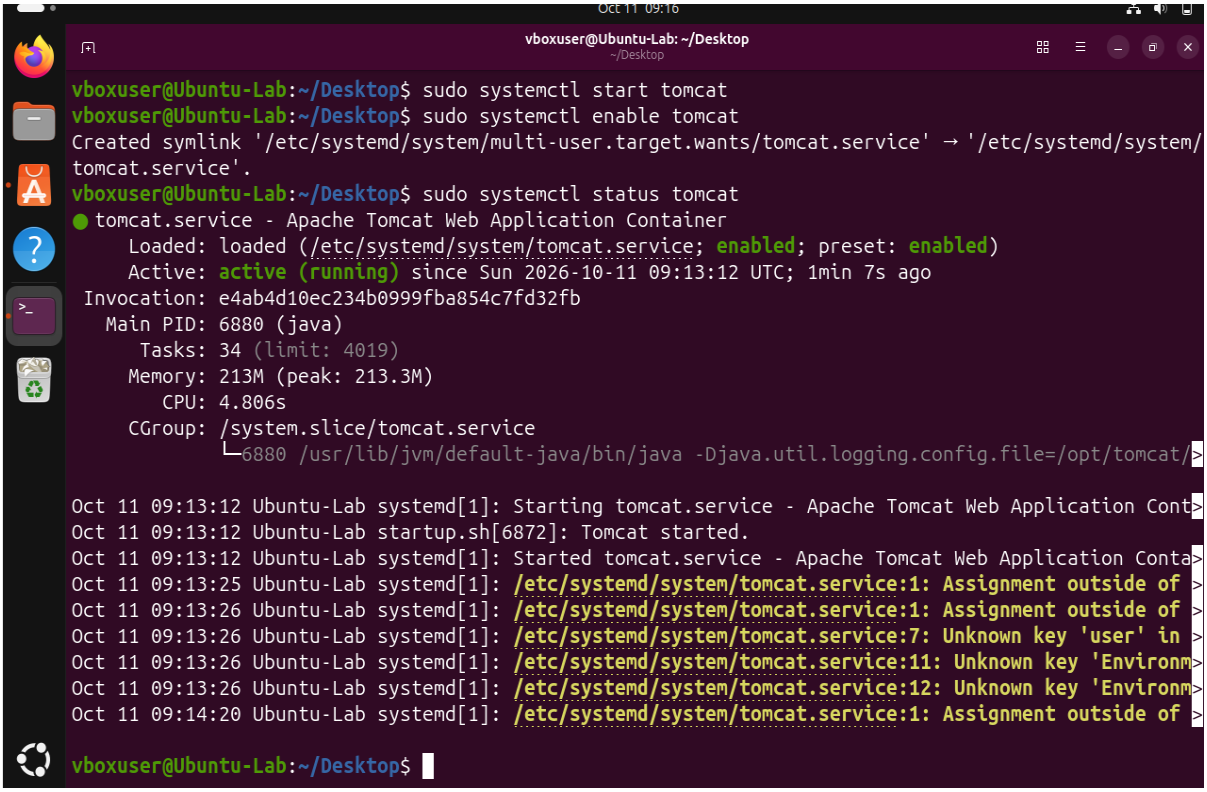
27. Configure and secure a MySQL Database.



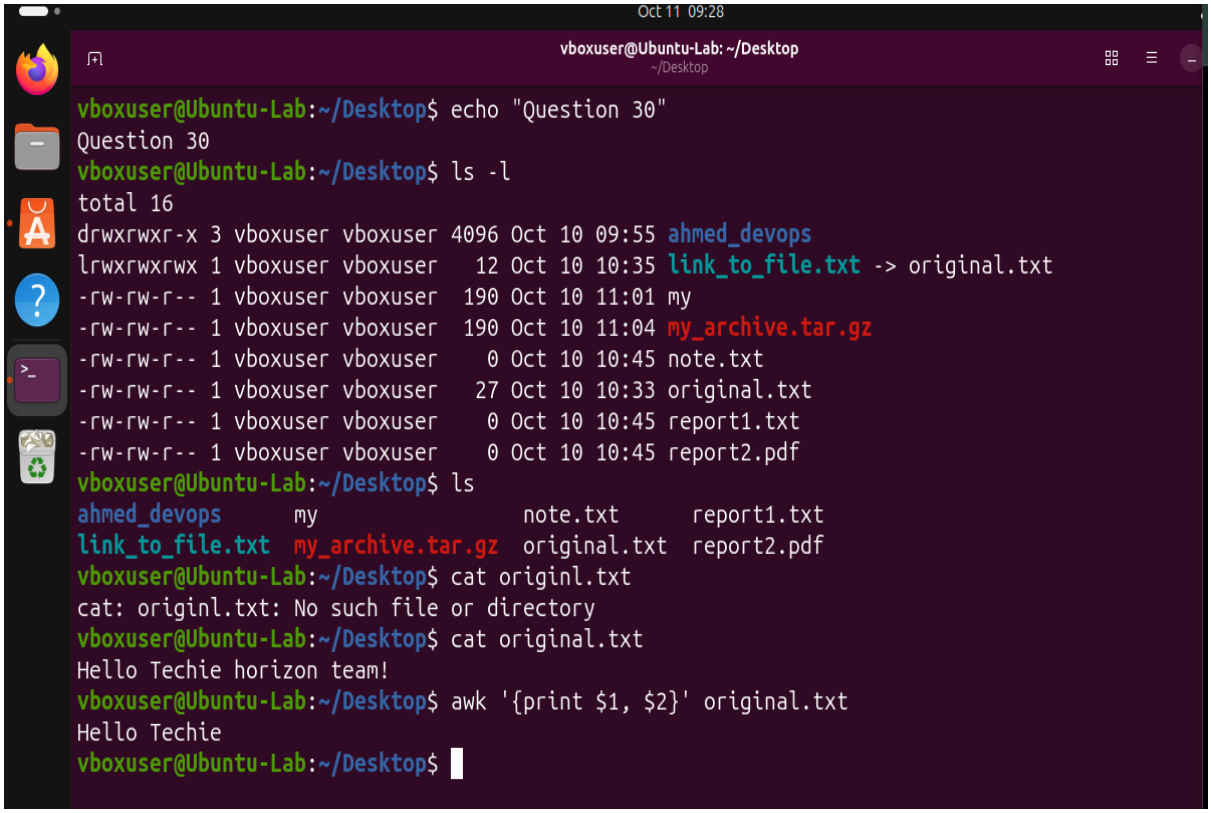
28. Set up a Application Server (e.g.,Apache Tomcat)



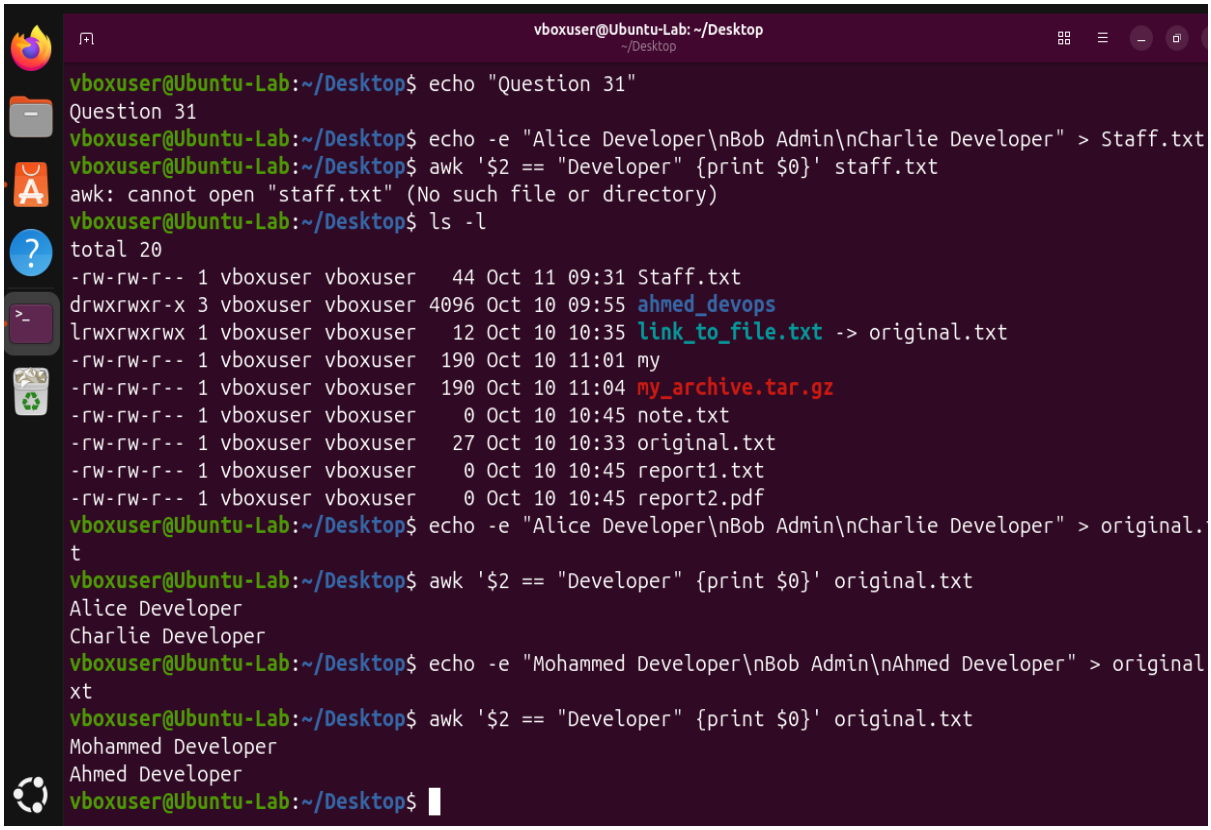
29. create a service file for Apache Tomcat.(Should execute by using systemtctl command).



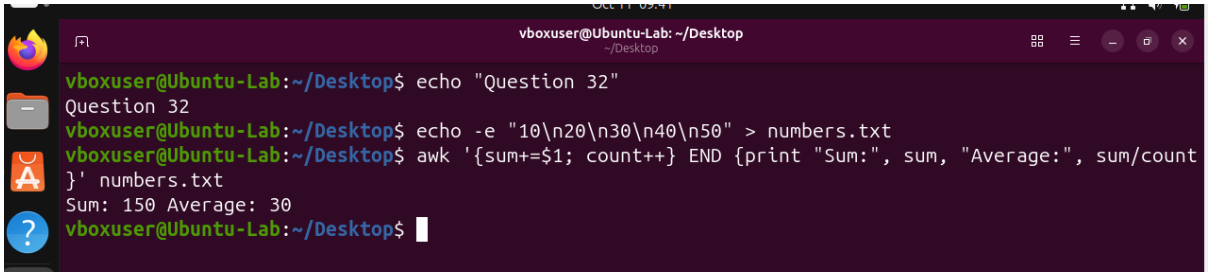
30.Print specific columns from a delimited file.



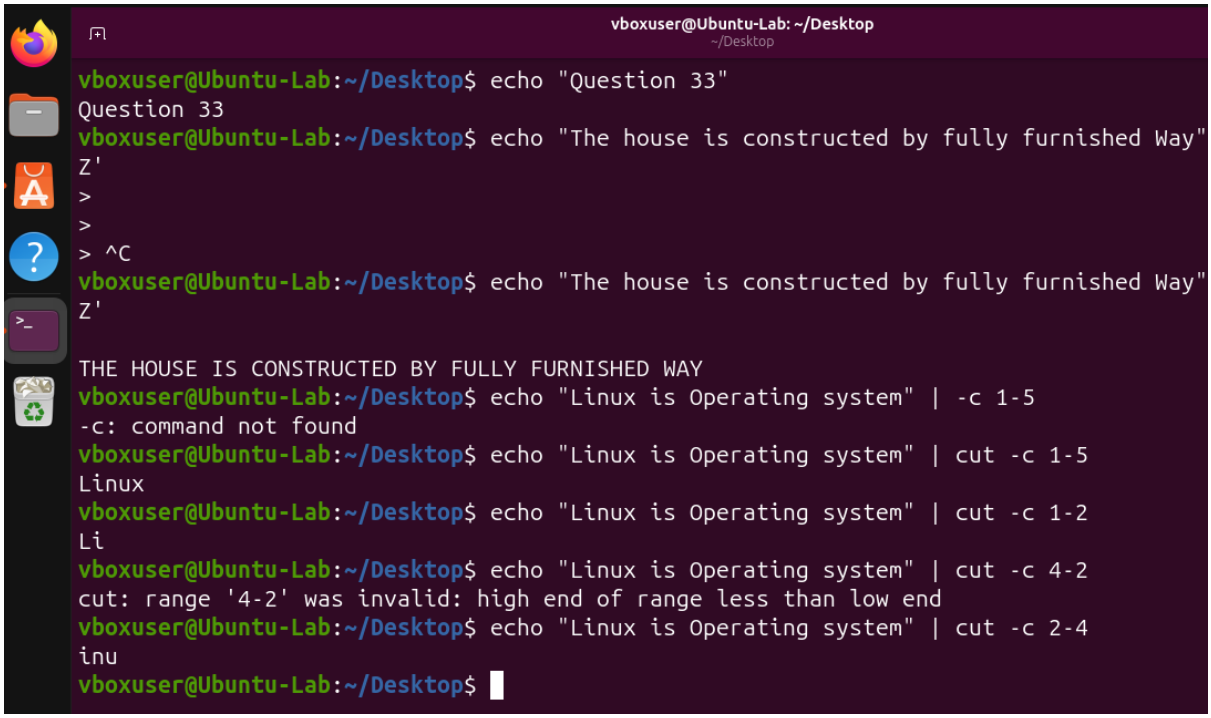
31. Filter and print lines based on a specific pattern or condition.



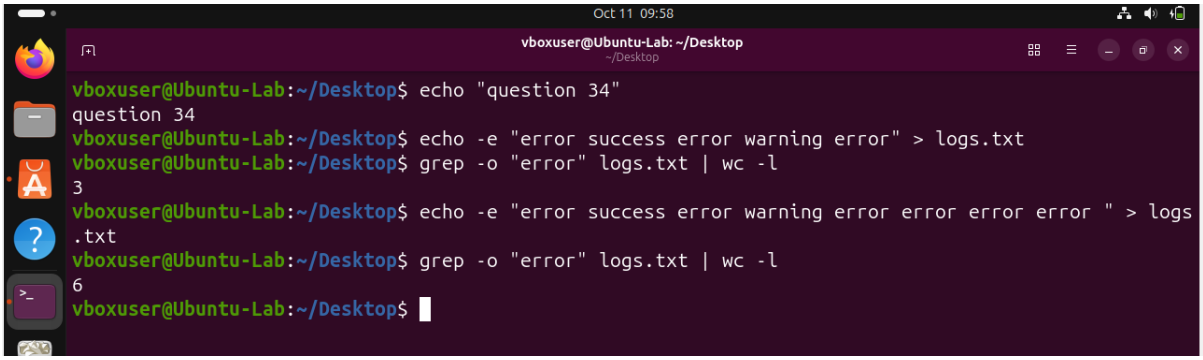
32. Calculate and print the average, sum, or other statistics of a column.



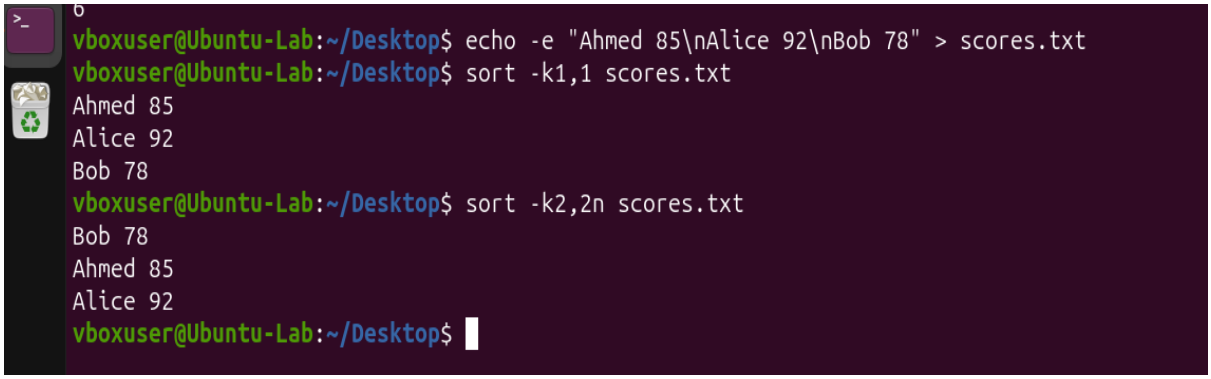
33. Perform string manipulation, such as extracting substrings or changing case.



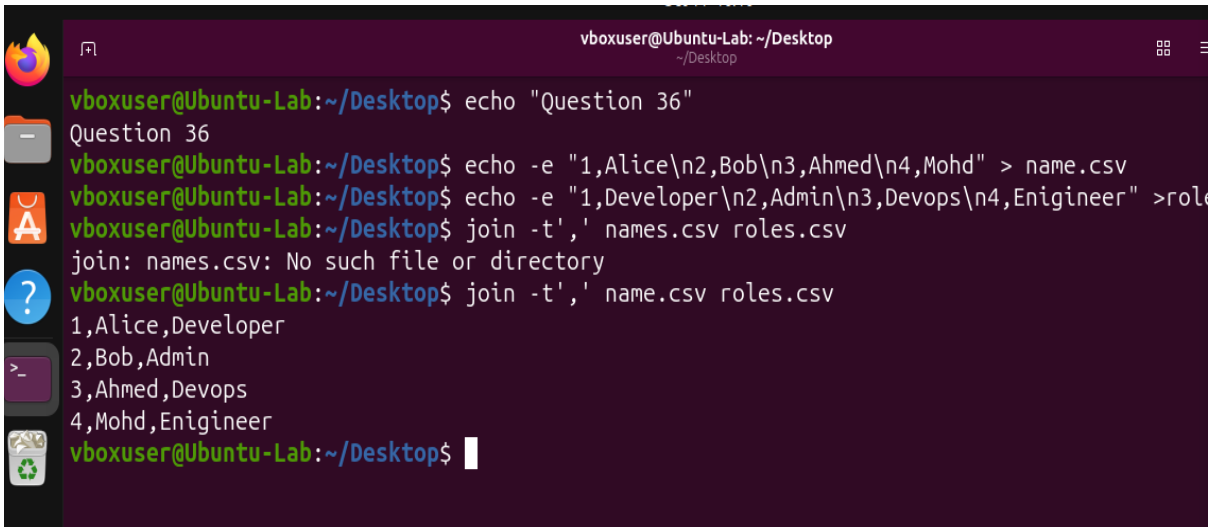
34. Count the occurrences of a specific pattern in a file.



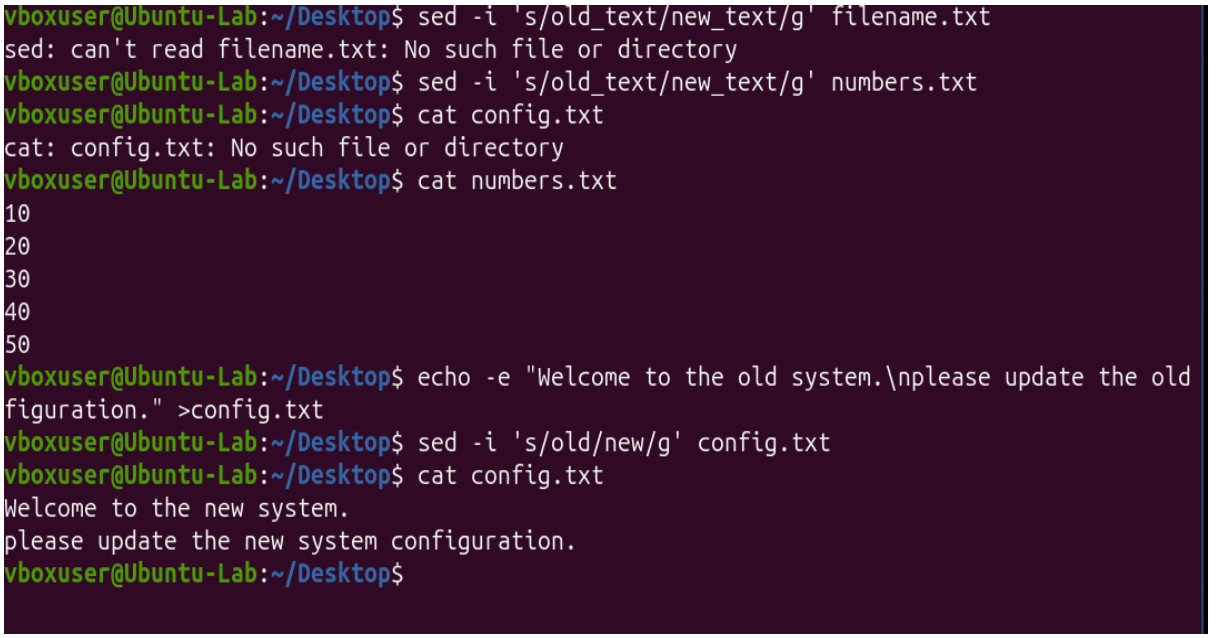
35. Sort lines based on a specific field or column.



36. Merge multiple files based on a common field or column.



37. Substitute text in a file using search and replace.



38. Delete specific lines based on a pattern or line number.

```
vboxuser@Ubuntu-Lab:~/Desktop$
vboxuser@Ubuntu-Lab:~/Desktop$ echo -e "apple\nbanana\nerror_line\ncherry\ndate" > items.t
vboxuser@Ubuntu-Lab:~/Desktop$ sed '/error/d' items.txt
apple
banana
cherry
date
vboxuser@Ubuntu-Lab:~/Desktop$ sed '2d' items.txt
apple
error_line
cherry
date
vboxuser@Ubuntu-Lab:~/Desktop$ sed '2,4d' items.txt
apple
date
vboxuser@Ubuntu-Lab:~/Desktop$
```

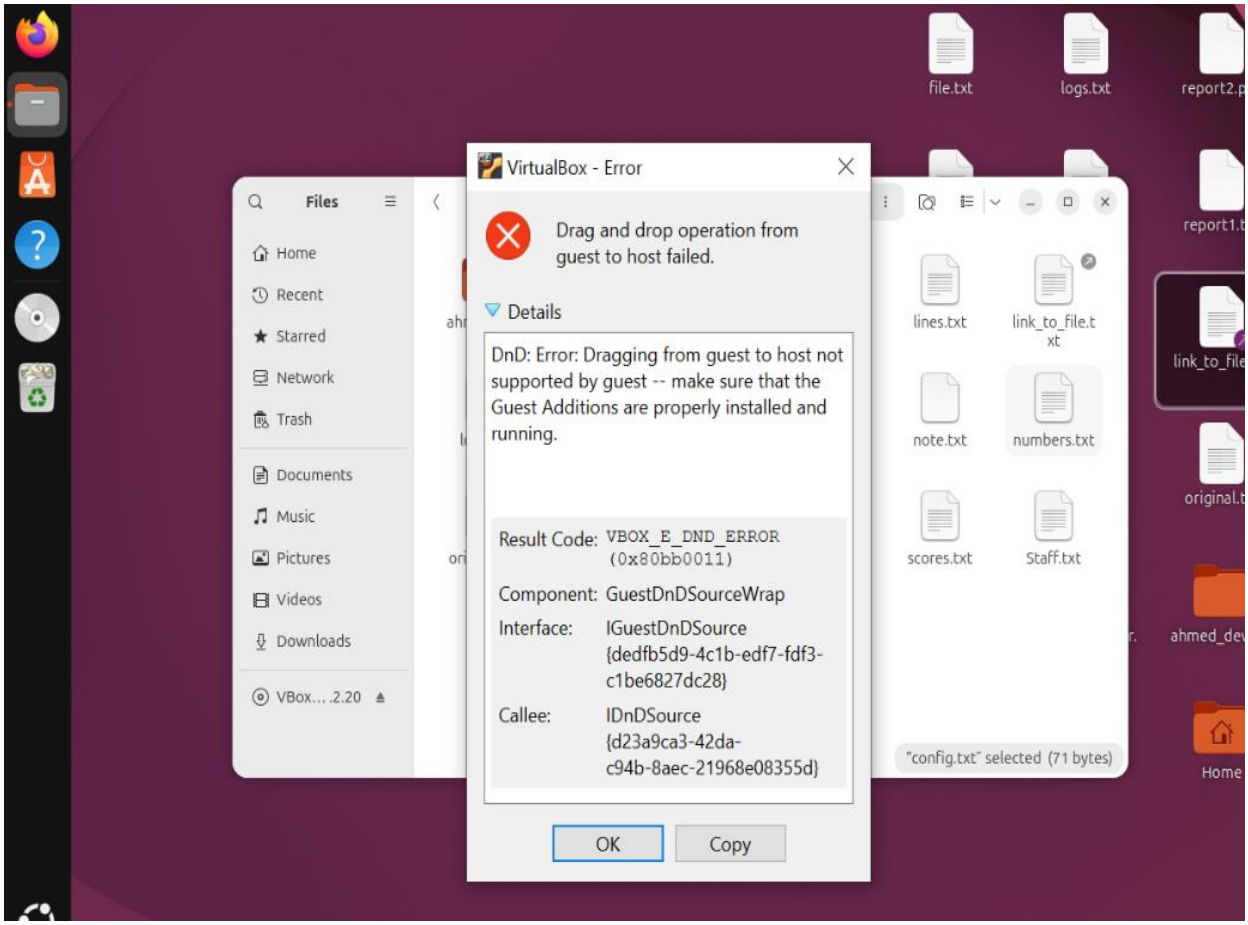
39. Append or insert text before or after a specific pattern or line.

```
vboxuser@Ubuntu-Lab:~/Desktop$ echo -e "line1\ntarget_line\nline3" > file.txt
vboxuser@Ubuntu-Lab:~/Desktop$ sed '/target_line/i\Inserted Before Text' file.txt
line1
Inserted Before Text
target_line
line3
vboxuser@Ubuntu-Lab:~/Desktop$ sed '/target_line/a\Appended After Text' file.txt
line1
target_line
Appended After Text
line3
vboxuser@Ubuntu-Lab:~/Desktop$
```

40. Print only specific lines from a file.

```
Appended After Text
line3
vboxuser@Ubuntu-Lab:~/Desktop$ echo -e "line1: Apple\nline2: Banana\nline3: Cherry\nline4: Date" > l
ines.txt
vboxuser@Ubuntu-Lab:~/Desktop$ sed -n '2p' lines.txt
line2: Banana
vboxuser@Ubuntu-Lab:~/Desktop$ sed -n '2,4p' lines.txt
line2: Banana
line3: Cherry
line4: Date
vboxuser@Ubuntu-Lab:~/Desktop$ awl 'NR==2 || NR==4' lnes.txt
Command 'awl' not found, did you mean:
  command 'aws' from snap aws-cli (1.45.46)
  command 'awk' from deb gawk (1:5.3.2-1ubuntu1)
  command 'awk' from deb mawk (1.3.4.20260129-1)
  command 'awk' from deb original-awk (2025-08-04-1)
  command 'axl' from deb afnix (3.8.0-1.1ubuntu1)
  command 'wwl' from deb wwl (1.3+db-3build1)
  command 'al' from deb mono-devel (6.14.1+ds2-2)
  command 'aws' from deb awscli (2.31.35-1)
See 'snap info <snapname>' for additional versions.
vboxuser@Ubuntu-Lab:~/Desktop$ awk 'NR==2 || NR==4' lnes.txt
awk: cannot open "lnes.txt" (No such file or directory)
vboxuser@Ubuntu-Lab:~/Desktop$ awk 'NR==2 || NR==4' lines.txt
line2: Banana
line4: Date
vboxuser@Ubuntu-Lab:~/Desktop$
```

41. Copy file from Linux to windows machine



I have tried to move it but file is not moving with the help of Google Gemini is locating to sign in to Gmail account in Morilla Firefox on ubuntu virtual box and share the file via mail.

42. 5 use cases for AWK and 5 use cases for sed

```
vboxuser@Ubuntu-Lab:~/Desktop$ echo -e "ID Name Department Salary\n101 Ahmed IT 60000\n102 MD HR 45000\n103 Shaik Finance 75000\n104 Sayed IT 5000" > data.txt
vboxuser@Ubuntu-Lab:~/Desktop$ echo "=== Orginal File ==="
=== Orginal File ===
vboxuser@Ubuntu-Lab:~/Desktop$ cat data.txt
ID Name Department Salary
101 Ahmed IT 60000
102 MD HR 45000
103 Shaik Finance 75000
104 Sayed IT 5000
vboxuser@Ubuntu-Lab:~/Desktop$ echo -e\n== AWK Example: print name Salary Columns ==="
> ^C
vboxuser@Ubuntu-Lab:~/Desktop$ echo -e "\n== AWK Example: print name Salary Columns ==="

== AWK Example: print name Salary Columns ===
vboxuser@Ubuntu-Lab:~/Desktop$ awk '{$2, $4}' data.txt
awk: line 1: syntax error at or near ,
vboxuser@Ubuntu-Lab:~/Desktop$ awk '{print $2, $4}' data.txt
Name Salary
Ahmed 60000
MD 45000
Shaik 75000
Sayed 5000
vboxuser@Ubuntu-Lab:~/Desktop$ echo -e "\n=== AWK example: Filter employees with salary > 50000 ==="

=== AWK example: Filter employees with salary > 50000 ===
vboxuser@Ubuntu-Lab:~/Desktop$ awk '$4 > 5000 {print $1, $2, $4}' data.txt
ID Name Salary
101 Ahmed 60000
102 MD 45000
103 Shaik 75000
vboxuser@Ubuntu-Lab:~/Desktop$ echo -e "\n=== SED Example: Replace IT with Engineering ==="

=== SED Example: Replace IT with Engineering ===
vboxuser@Ubuntu-Lab:~/Desktop$ sed 's/IT/Engineering/g' data.txt
ID Name Department Salary
101 Ahmed Engineering 60000
102 MD HR 45000
103 Shaik Finance 75000
104 Sayed Engineering 5000
vboxuser@Ubuntu-Lab:~/Desktop$ echo -e "\n=== SED Example: print lines 2 through 3 only ==="

=== SED Example: print lines 2 through 3 only ===
vboxuser@Ubuntu-Lab:~/Desktop$ sed -n '2,3p' data.txt
101 Ahmed IT 60000
102 MD HR 45000
```